Cascade Control with Reinforcement Learning for an Industrial Exoskeleton

Andrey Yatsun
Mechatronics and Robotics Department
South-West State University
Kursk, Russia
ayatsun@ya.ru

Gavriil Fursov
Mechatronics and Robotics Department
South-West State University
Kursk, Russia
from.skveki1@mail.ru

Ekaterina Saveleva
Mechatronics and Robotics Department
South-West State University
Kursk, Russia
katyasavekkursk@yandex.ru

2026 International Russian Automation Conference (RusAutoCon) | 979-8-3195-0830-0/26/$31.00 ©2026 IEEE | DOI: 10.1109/RUSAUTOCON69876.2026.11707381

***Abstract*—This paper presents an adaptive control framework for an industrial lower-limb exoskeleton based on cascade control combined with reinforcement learning (RL). The multilevel hierarchical structure of the cascade controller effectively decouples the fast inner-loop current and torque dynamics from the slower outer-loop kinematic tracking tasks. This decomposition significantly enhances overall system bandwidth while mitigating undesirable dynamic coupling between adjacent joints during complex gait cycles. The proposed approach aims to ensure safe and intuitive physical human–robot interaction by dynamically compensating for parametric uncertainties, while simultaneously improving adaptability to heterogeneous individual biomechanical characteristics and non-stationary task conditions. A hybrid actuation architecture with electric drives and gravity compensation is considered. Cascade parameters are adjusted online using policy-gradient-based reinforcement learning to optimize interaction forces, trajectory tracking accuracy, and energy efficiency. Experimental results obtained on a laboratory prototype demonstrate improved stability, comfort, and reduced muscular load compared to fixed-parameter cascade control.**

Keywords—industrial exoskeleton, cascade control, reinforcement learning, human–robot interaction, biomechatronic systems

I. Introduction

Industrial exoskeletons represent a rapidly developing class of biomechatronic systems designed to augment human physical capabilities, reduce musculoskeletal load, and prevent occupational injuries in manufacturing, logistics, construction, and maintenance tasks [1-4]. Industrial exoskeletal systems must operate under highly variable external conditions, including unpredictable user motion, changing payloads, and prolonged duty cycles [3,4] . These requirements impose strict constraints on control system design, particularly in terms of safety, robustness, and adaptability. Early generations of industrial exoskeletons relied predominantly on passive mechanical structures, including spring-based or gravity-compensated mechanisms, aimed at redistributing loads without active actuation. While such solutions provide high reliability and low energy consumption, they offer limited adaptability and cannot actively assist complex dynamic motions. The transition toward active and semi-active exoskeletons has therefore motivated the development of advanced control strategies capable of managing direct physical human–robot interaction. Conventional position or torque control methods, widely used in industrial robotics, are poorly suited for wearable systems due to their inherent rigidity. Excessive tracking accuracy in the presence of human interaction may lead to uncomfortable or even unsafe force levels. As a result, cascade control has become a dominant paradigm for exoskeleton applications. By regulating the dynamic relationship between motion and force, cascade control enables compliant behavior and natural interaction patterns. However, classical cascade control typically relies on fixed stiffness and damping parameters, selected empirically during system tuning [5] . In industrial scenarios, where operator anthropometry, fatigue level, and task dynamics vary significantly, fixed parameters cannot provide optimal performance across all conditions. This limitation has motivated research into adaptive and learning-based control strategies. Recent studies have explored adaptive cascade control, model-based optimization, and machine learning techniques for exoskeleton control. Reinforcement learning, in particular, has demonstrated strong potential for handling complex, nonlinear systems with uncertain dynamics. RL allows the control system to improve performance through interaction experience, without requiring explicit modeling of human biomechanics.

Despite this potential, the integration of RL into low-level physical interaction control for industrial exoskeletons remains insufficiently explored. This paper addresses this gap by proposing an cascade control framework augmented with reinforcement learning for an industrial lower-limb exoskeleton. The contribution of the work includes:

- a kinematic and drive-level model of an exoskeleton with two active hip joints,
- a structured adaptive cascade control algorithm based on policy-gradient reinforcement learning,
- practical recommendations for hardware implementation on modern embedded control platforms.

II. System Description and Kinematic Model

A single-link model of the lower limb exoskeleton is considered, corresponding to the hip joint architecture implemented in the laboratory prototype (Fig. 1).

The considered exoskeleton is a lower-limb industrial assistive device with two active degrees of freedom, corresponding to flexion–extension movements in the left and right hip joints. Knee and ankle joints are passive and mechanically aligned with the operator's anatomical joints. Such

2026 International Russian Automation Conference (RusAutoCon)

979-8-3195-0830-0/26/$31.00 ©2026 IEEE

Authorized licensed use limited to: Lomonosov Moscow State University. Downloaded on October 12,2026 at 07:12:57 UTC from IEEE Xplore. Restrictions apply.

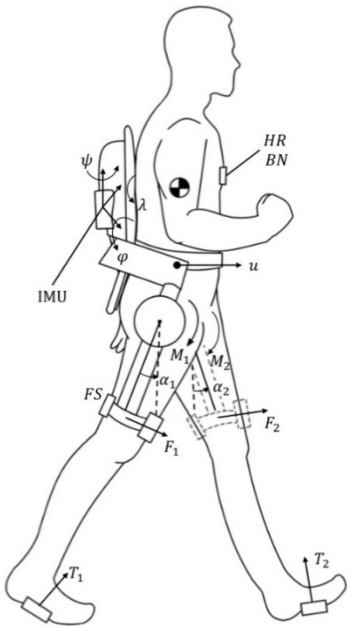


Fig. 1. Scheme of lower-limb industrial assistive exoskeleton. T_1, T_2- planar pressure forces; F_1, F_2- the interaction forces between the operator and the hip joint ; u- operator velocity; α1, α2- hip joint angles; M_1, M_2-moment torgues on the hip joints; vector; λ; φ; ψ- rotation angles above sagittal, frontal and vertical axes; FS-force sensor ; HR-heart rate ; BN- breath number ; IMU-inertial measurement unit .

a configuration provides a balance between system complexity, weight, and assistive capability [6].

Electric drives are installed at the hip joints, while gravity compensation mechanisms reduce static load on the actuators. The system is equipped with joint angle sensors, force–torque sensors, inertial measurement units (IMUs), and plantar pressure sensors. Each active hip joint is driven by a permanent magnet synchronous motor (PMSM) coupled with a reduction gearbox. PMSM drives are widely used in exoskeletons due to their high power density, efficiency, and controllability.

In the dq-reference frame, the electrical dynamics of the PMSM are described by [7]:

$$v_d = Ri_d + L_d\frac{di_d}{dt} - \omega_e L_q i_q$$

$$v_q = Ri_q + L_q\frac{di_q}{dt} + \omega_e(L_d i_d + \psi_f), \qquad (1)$$

where i_d , i_q are the stator currents, v_d , v_q are the applied voltages, R is the stator resistance, L_d and L_q are inductances, ω_e is the electrical angular velocity, and ψ_f is the permanent magnet flux linkage.

The electromagnetic torque is given by:

$$T_e = \frac{3}{2}p\left[\psi_f i_q + (L_d - L_q)i_d i_q\right], \qquad (2)$$

where p is the number of pole pairs. In the proposed system, field-oriented control (FOC) is employed, enabling decoupled torque and flux regulation. The mechanical dynamics of the joint are described by:

$$T_e = J\ddot{q} + B\dot{q} + \tau_{load}, \qquad (3)$$

where J is the equivalent inertia, B is viscous friction, and τ_{load} represents external interaction torques.

III. Cascade control System architectrute

Control strategies for industrial exoskeletons can be broadly classified into position-based, force-based, and interaction-oriented approaches. Position-based control aims to strictly track predefined joint trajectories. While effective for repetitive tasks, this approach lacks compliance and is sensitive to modeling errors. Force and torque control methods attempt to regulate interaction forces directly, often requiring accurate force sensing and dynamic models. In wearable systems, sensor noise, delays, and variability of human behavior significantly complicate pure force control. Interaction-oriented approaches, including cascade and admittance control, offer a compromise by shaping the dynamic behavior of the system rather than enforcing exact motion or force values. These methods are particularly suitable for exoskeletons, as they allow the human operator to remain the primary source of motion while the device provides assistive or resistive forces. In industrial exoskeletons, cascade control is commonly combined with gravity compensation, load estimation, and high-level task recognition. However, most commercial and experimental systems employ static cascade parameters, leading to suboptimal performance when operating conditions change. Adaptive and learning-based extensions are therefore a natural evolution of existing control architectures.

The control architecture consists of three layers (Fig. 2):

1. sensor data acquisition and state estimation;

2. cascade control loop;

3. reinforcement learning module for adaptive control [8-10].

Reinforcement learning has been applied to robotic locomotion, manipulation, and exoskeleton control, including Q-

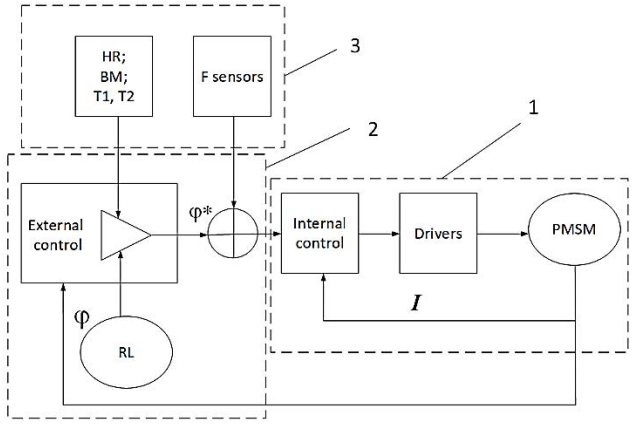


Fig. 2. Structural scheme of the digital control system with RL. 1- internal control contoure; 2- external control contoure; 3- external influences; φ- link rotation angle; φ*- desired link rotation angle; *I*- amperage; FS-force sensor ; HR-heart rate ; BN- breath number; reinforcement learning (RL).

Authorized licensed use limited to: Lomonosov Moscow State University. Downloaded on October 12,2026 at 07:12:57 UTC from IEEE Xplore. Restrictions apply.

learning and policy gradient methods. However, many existing approaches focus on high-level gait pattern generation rather than low-level physical interaction. In contrast, the present work integrates RL directly into the cascade control loop, allowing continuous adaptation of stiffness and damping parameters based on measured interaction forces, kinematics, and energy consumption. Below we will consider the Algorithm for tuning the PID controller using reinforcement learning

This algorithm includes steps 1-4 and describes how a trainable agent (a neural network) tunes the coefficients of classical PID controllers in real-time to improve the control quality of a complex system (e.g., an actuator or robot).

1. Data Preparation (State Vector Formation)

The agent does not directly control current or voltage. It receives information about the state of the system as a state vector (s_t).To make the neural network work effectively, all data is scaled (normalized) into a common range, approximately from -1 to 1. This ensures that no single parameter "drowns out" the others and speeds up learning.

The state vector consists of 9 measurements. Each measurement is divided by its own constant (scale) (Table 1):

TABLE I. THE STATE VECTOR MEASUREMENTS

№	Parameter	Description	Scaling Constant	Range
1	e_θ	Position error (radians)	0.5	[-1, 1]
2	$\dot{e}_\theta$	Derivative of position error (rad/s)	5	[-1, 1]
3	e_ω	Speed loop error (rad/s)	5	[-1, 1]
4	τ_{int}	Interaction torque / resistance (N·m)	50	[-0.5, 0.5]
5	E	Accumulated energy (Joules)	200	[-1, 1]
6	θ_{act}	Current joint angle (radians)	1	[-1, 1]
7	ω_{act}	Current angular velocity (rad/s)	10	[-1, 1]
8	τ_{ref_curr}	Desired/reference torque (load on drive)	200	[-1, 1]
9	$stage$	Gait phase (step cycle stage: 0, 1, 2, or 3)	3	[0, 1]

If values are left in their original units (e.g., meters and newton-meters), large numbers will dominate small ones. The neural network would react only to the largest changes, ignoring important details. Normalization equalizes their contribution.

2. Action Generation and Coefficient Update

Based on the received state vector, the neural network computes an action (a_t). These are increments to be added to the base controller coefficients. This action is performed through the following steps *A*-*C*.

A. Action Computation

The neural network outputs a vector of 5 numbers ($a_t \in R^5$). Each number corresponds to a change in one of the following coefficients:

$K_{p_{pos}}$ — Proportional gain for the position loop.

$K_{i_{pos}}$ — Integral gain for the position loop.

$K_{d_{pos}}$ — Derivative gain for the position loop.

$K_{p_{vel}}$ — Proportional gain for the speed loop.

$K_{i_{vel}}$ — Proportional gain for the speed loop.

$K_{d_{vel}}$— Integral gain for the speed loop.

To prevent the agent from getting stuck with one set of parameters and to encourage exploration of new solutions, Gaussian noise is added to the result.

B. Action Scaling

The output values of the neural network are clipped to the range [-1.0, 1.0]. Then, these increments are scaled using a multiplier of 1.5. This means the coefficients can change by ±150% of their base value.

C. Coefficient Update (Smoothing)

Coefficients do not change instantly. Instantaneous jumps can lead to system instability. Exponential smoothing is applied:

$$K(k) = (1 - \beta_{rl}) \cdot K(k-1) + \beta_{rl} \cdot K_{target}\ , \qquad (4)$$

where:

$K(k)$ — New coefficient at the current step.

$K(k-1)$ — Coefficient at the previous step.

$\beta_{rl} = 0.25$ — Smoothing factor. These guarantees smooth parameter changes: they respond quickly enough to gait phase changes but avoid sharp spikes.

3. Reward Function and Agent Training

The agent learns through trial and error. Its goal is to maximize the total reward (R).

Reward Formula (r_t) At each time step, a penalty for errors is calculated:

$$r_t = -w_1 e_\theta^2 - w_2 \tau_{int}^2 - w_3 E\ , \qquad (5)$$

where weights are:

$w_1 = 8.0$ (importance of position accuracy).

$w_2 = 3.0$ (importance of reducing interaction force/comfort).

$w_3 = 0.5$ (importance of energy efficiency). A high weight w_2 forces the agent to primarily reduce the interaction torque (τ_{int}), even if it slightly worsens tracking accuracy. This is done for operator comfort or mechanism safety.

Training Process (Backpropagation) At the end of each run (episode), the neural network analyzes all its actions:

Advantage Calculation: The sum of future rewards is compared to the critic's prediction.

Authorized licensed use limited to: Lomonosov Moscow State University. Downloaded on October 12,2026 at 07:12:57 UTC from IEEE Xplore. Restrictions apply.

Weight Update: The network adjusts its internal parameters so that in similar situations, it chooses actions that brought more reward.

Actor learning rate: $\alpha_{actor} = 0.0005$

Critic learning rate: $\alpha_{critic} = 0.001$

4. Frequency Synchronization

For correct system operation, timing intervals must be observed:

- The RL module operates at 500 Hz. It makes decisions about changing coefficients very quickly.
- The position loop operates at 200 Hz. It receives new coefficients from the RL module and controls the motion.

This separation is critical for safety. The RL agent works slowly compared to electrical processes inside the drive (which can occur at frequencies up to 10 kHz). The RL module does not interfere directly with high-frequency electronics; instead, it sets the "rules of the game" for the lower-level control, ensuring determinism and predictability of the system behavior.

IV. Cascade Control and Adaptive Tuning Modelling

Cascade control defines a desired dynamic relationship between joint motion and interaction torque [11]:

$$\tau_{int} = M_d(\ddot{q} - \ddot{q}_d) + B_d(\dot{q} - \dot{q}_d) + K_d(q - q_d) \quad (6)$$

Here, K_d and B_d are the virtual stiffness and damping matrices. In classical implementations, these parameters are constant.

In the proposed approach, K_d and B_d are adapted online using reinforcement learning. The system state includes joint angles, velocities, interaction torques, and estimated user effort. The action vector corresponds to incremental changes in stiffness and damping coefficients.

A policy-gradient reinforcement learning algorithm is used to update the control policy. The reward function penalizes large interaction forces, tracking errors, and excessive energy consumption. This formulation encourages compliant yet effective assistance [12].

The adaptive tuning algorithm operates in parallel with the low-level motor control loop, ensuring real-time performance and safety constraints.

Cascade control regulates the dynamic relationship between joint motion and interaction forces according to a virtual mass–spring–damper model. For a single joint, the control law can be expressed as:

$$\tau_{int} = M_d\left(\ddot{\theta} - \ddot{\theta}_d\right) + B_d\left(\dot{\theta} - \dot{\theta}_d\right) + K_d(\theta - \theta_d), \quad (7)$$

where θ is the joint angle, θ_d is the desired trajectory, K_d and B_d are the stiffness and damping coefficients, and τ_{int} represents the interaction torque.

The actuator torque command is computed as:

$$\tau = K_p(\theta_d - \theta) + K_d\left(\dot{\theta}_d - \theta\right) + \tau_{comp}, \quad (8)$$

where τ_{comp} compensates gravitational and inertial effects. Proper tuning of K_p and K_d is critical for achieving both stability and comfort.

A policy gradient method is employed to directly optimize the control policy. The reward function is defined as [13-16]:

$$r_t = -\omega_1 e_\theta^2 - \omega_2 \tau_{int}^2 - \omega_3 E_t, \quad (9)$$

where e_θ is the trajectory tracking error, τ_{int} is the interaction torque, and E_t is the estimated energy consumption. This formulation encourages accurate motion tracking, low interaction forces, and energy efficiency.

The policy parameters are updated according to:

$$\nabla_\theta J = E[r_t \nabla_\theta log\ \pi(a_t \mid s_t; \theta)]. \quad (10)$$

Here $\nabla_\theta J$ is the gradient of the objective function with respect to the policy parameters θ. It indicates the direction in which the parameters should be changed to improve the policy's performance. E is the mathematical expectation. The formula describes the average value over all possible trajectories or over the data distribution. r_t is the reward received at step t. $\pi(a_t \mid s_t; \theta)$ is the probability of choosing action a_t in state s_t given the parameters θ. $\nabla_\theta \log\ \pi(a_t \mid s_t; \theta)$ is the gradient of the logarithm of the action probability. This technique is used to improve the stability and efficiency of learning.

The proposed control architecture is suitable for implementation on modern 32-bit microcontrollers or system-on-chip platforms with floating-point support. Typical candidates include ARM Cortex-M7 or Cortex-A series processors.

The PMSM control loop, including current regulation and FOC, is executed at high frequency (10–20 kHz), while the cascade control and RL-based adaptation operate at lower rates (500–1000 Hz). Sensor data acquisition is synchronized using DMA and hardware timers to minimize latency.

The exoskeleton modelling parameters are are given in Table 2.

Safety mechanisms are implemented at both hardware and software levels, including torque limiting, watchdog timers, and emergency stop handling.

- Joint angle: from −1 to −0.81 rad;
- Angular velocity: up to 36 rad/s;
- Torque: 110 N·m (34% of maximum);
- Motor current: 45 A (45% of maximum).

To adapt cascade parameters online, a reinforcement learning agent is introduced. The state vector includes joint angles, angular velocities, interaction torques, and estimated muscle activation indicators. The action vector consists of incremental adjustments to stiffness and damping coefficients.

Authorized licensed use limited to: Lomonosov Moscow State University. Downloaded on October 12,2026 at 07:12:57 UTC from IEEE Xplore. Restrictions apply.

TABLE II. EXOSKELETON MODELING PARAMETERS

Engine and drive parameters (PMCM X12-320)	
Gear ratio of the gearbox	20
Gearbox efficiency coefficient	75%
Phase resistance	0.120 Om
Inductance	0.055 / 0.050 mHn
Flux linkage	0.00733 V*c
Number of pole pairs	20
Magnetic moment on the motor shaft	0.220 N*m/A
Magnetic moment at the gearbox output	3.30 N*m/A
Bus voltage	48. 0 V (working limit: 43.2 V)
Current: nominal / limit / maximum	30/ 45/ 100 A
Torque: nominal / limit / maximum	35.0 / 110.0 / 320.0 N*m
Biomechanical parameters	
Mass: Femur / Exoskeleton / Total	8.5/2.4/10.9 kg
Femur Length	0.43 m
Distance to the mass senter	0.181 m. (42% per lenght)
Moment of inertia of the load	0.4878 kg*m^2
Passive rigidity	0.5 N*m*c/rad
Passive damping	0.080 N*m*c/rad
Viscous friction of the drive	0.050 N*m*c/rad
Basic regulator coefficients	
Position contour coefficient	25/ 1.5/ 6.0
Velocity contour coefficient	1/0.6
Current loop coefficient	0.6/12
Frequencies and sampling	
Integration step	100 microsec. (10 kHz)
Current circuit (internal)	10000 Hz (each step)
Speed contour (medium)	1000 Hz (each 10 steps)
Position contour (external)	200 Hz (each 50 steps)
RL adaptation	500 Hz (each 20 steps)
Episode duration	15 sec. (total number of steps: 150001)

V. RESULTS AND FUTURE WORK

A matematical modelling was developed and tested performing load-handling tasks. The proposed adaptive cascade control was compared with fixed-parameter cascade control. The results of matematical modelling are developed on the graphs below (Fig. 3-5).

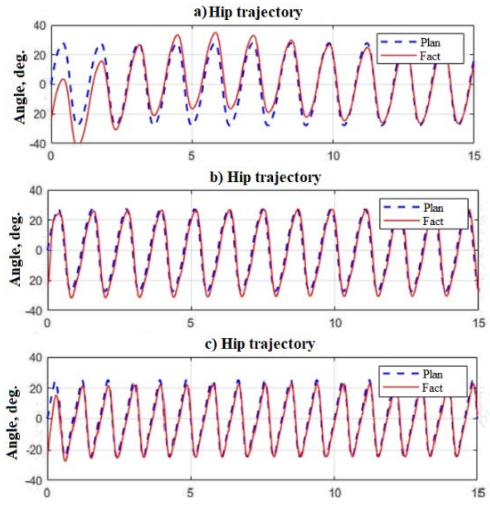


Fig. 3. Hip tragectory in different modes a) high speed; b) middle speed; c) low speed.

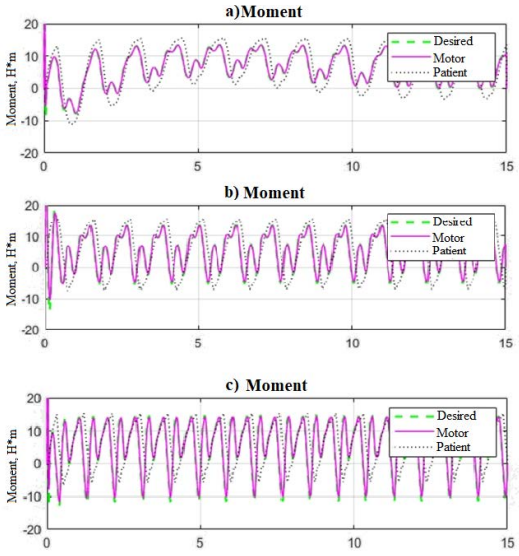


Fig. 4. Moment value in different modes a) high speed; b) middle speed; c) low speed.

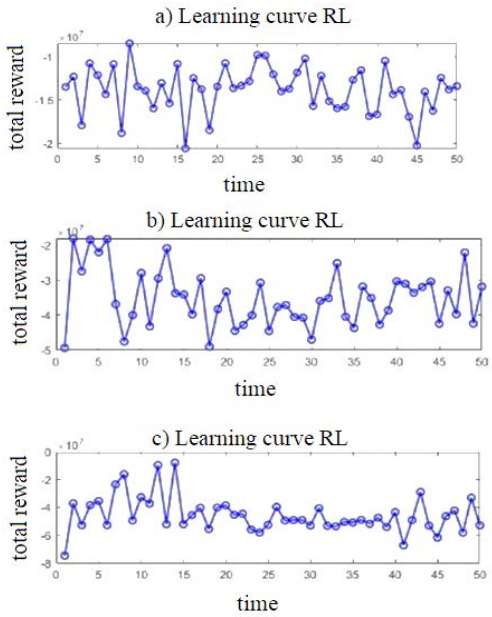


Fig. 5. Learning curve in different modes a) high speed; b) middle speed; c) low speed.

VI. CONCLUSION

Modelling results indicate that the RL-based approach reduces peak interaction forces by up to 18% and improves trajectory tracking accuracy by approximately 12%. Subjective feedback also confirmed improved comfort and smoother interaction. The obtained results demonstrate the potential of reinforcement learning for enhancing cascade control in industrial exoskeletons [17,18]. Unlike predefined parameter tuning, the proposed method adapts to individual users and task conditions in real time. However, training stability and safety constraints remain critical challenges and require further investigation [19,20]. Further experimental studies will be

Authorized licensed use limited to: Lomonosov Moscow State University. Downloaded on October 12,2026 at 07:12:57 UTC from IEEE Xplore. Restrictions apply.

conducted with healthy subjects participating in lifting tasks to confirm the correctness of the mathematical model.

ACKNOWLEDGMENT

This research was supported by the Russian Science Foundation under project No. 25-21-20104.

REFERENCES

[1] S. F. Jatsun, A. S. Pechurin, and E. V. Savelyeva, "Mathematical modeling of the interaction of elements in an electromechanical multi-link system on the example of a lower limb rehabilitation exoskeleton," (in Russian), Izvestiya Yugo-Zapadnogo Gosudarstvennogo Universiteta. Seriya: Upravleniye, Vychislitelnaya Tekhnika, Informatika. Meditsinskoye Priborostroyenie, vol. 15, no. 2, pp. 204–220, 2025.

[2] S. F. Jatsun, L. B. Kislyak, E. V. Savelyeva, and G. A. Fursov, "Models and algorithms for generating control setpoints for a two-link foot of a rehabilitation exoskeleton," (in Russian), Izvestiya Yugo-Zapadnogo Gosudarstvennogo Universiteta. Seriya: Upravleniye, Vychislitelnaya Tekhnika, Informatika. Meditsinskoye Priborostroyenie, vol. 14, no. 4, pp. 60–77, 2025.

[3] A. A. Knyazev, A. V. Fyodorov, and A. S. Yatsun, "Algorithm for controlling the movement of the actuating mechanism of a device for active-passive mechanotherapy of the ankle joint," (in Russian), Izvestiya Yugo-Zapadnogo Gosudarstvennogo Universiteta, vol. 26, no. 4, pp. 75–87, 2023.

[4] S. Jatsun, A. Malchikov, A. Yatsun, and E. Saveleva, "Studying of copying control system with nonlinear measurer," in Proc. 16th Int. Conf. Electromechanics and Robotics "Zavalishin's Readings" (ER(ZR) 2021), St. Petersburg, Russia, Apr. 14–17, 2021, Singapore: Springer, 2021, pp. 13–23.

[5] H. Shao, W. Hu, L. Yang, W. Wang, S. Suzuki, and Z. Gao, "Intelligent impedance strategy for force–motion control of robotic manipulators in unknown environments via expert-guided deep reinforcement learning," Processes, vol. 13, no. 8, 2025, Art. no. 2526, doi: 10.3390/pr13082526.

[6] M. Sharifi, S. Tripathi, Y. Chen, Q. Zhang, and M. Tavakoli, "Reinforcement learning methods for assistive and rehabilitation robotic systems: A survey," IEEE Transactions on Systems, Man, and Cybernetics: Systems, 2025, doi: 10.1109/TSMC.2025.3553142.

[7] M. Akrami, "Design and implementation of electric motor drive for automotive cooling water pump," Ph.D. dissertation, Univ. Strasbourg, France, 2023.

[8] C. Tang, F. Abbatematteo, and P. Stone, "Deep reinforcement learning for robotics: A survey of real-world successes," Annual Review of Control, Robotics, and Autonomous Systems, vol. 8, no. 1, pp. 153–188, 2025, doi: 10.1146/annurev-control-012824-013215.

[9] M. Aljamal, S. Patel, and A. Mahmood, "Comprehensive review of robotics operating system-based reinforcement learning in robotics," Applied Sciences, vol. 15, no. 4, 2025, Art. no. 1840, doi: 10.3390/app15041840.

[10] L. Smith et al., "Learning and adapting agile locomotion skills by transferring experience," 2023, arXiv:2304.09834. [Online]. Available: https://arxiv.org/abs/2304.09834

[11] F. J. Abu-Dakka and M. Saveriano, "Variable impedance control and learning—A review," Frontiers in Robotics and AI, vol. 7, 2020, Art. no. 590681, doi: 10.3389/frobt.2020.590681.

[12] E. H. Sumiea, S. J. Abdulkadir, H. S. Alhussian, S. M. Al-Selwi, A. Alqushaibi, M. G. Ragab, and S. M. Fati, "Deep deterministic policy gradient algorithm: A systematic review," Heliyon, vol. 10, no. 9, 2024, doi: 10.1016/j.heliyon.2024.e33256.

[13] C. Huang, G. Wang, Z. Zhou, R. Zhang, and L. Lin, "Reward-adaptive reinforcement learning: Dynamic policy gradient optimization for bipedal locomotion," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 45, no. 6, pp. 7686–7695, 2022, doi: 10.1109/TPAMI.2022.3191845.

[14] Y. Wang and S. Zou, "Policy gradient method for robust reinforcement learning," in Proc. Int. Conf. Machine Learning (ICML), PMLR, 2022, pp. 23484–23526.

[15] J. Zhang, A. Koppel, A. S. Bedi, C. Szepesvari, and M. Wang, "Variational policy gradient method for reinforcement learning with general utilities," in Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020, pp. 4572–4583.

[16] T. Chen, K. Zhang, G. B. Giannakis, and T. Başar, "Communication-efficient policy gradient methods for distributed reinforcement learning," IEEE Transactions on Control of Network Systems, vol. 9, no. 2, pp. 917–929, 2021, doi: 10.1109/TCNS.2021.3118012.

[17] L. He, Z. Li, Y. Dong, and T. Zhao, "Dynamic event-driven composite optimal impedance control for lower limb exoskeletons using finite-time reinforcement learning," Expert Systems with Applications, vol. 288, 2025, Art. no. 128181, doi: 10.1016/j.eswa.2025.128181.

[18] Y. Sun, Z. Peng, J. Hu, and B. K. Ghosh, "Event-triggered critic learning impedance control of lower limb exoskeleton robots in interactive environments," Neurocomputing, vol. 564, 2024, Art. no. 126963, doi: 10.1016/j.neucom.2024.126963.

[19] A. Giallanza, A. Portioli-Staudacher, G. Boccuzzo, and L. Mangili, "Occupational health and safety issues in human-robot collaboration: State of the art and open challenges," Safety Science, vol. 169, 2024, Art. no. 106313, doi: 10.1016/j.ssci.2024.106313.

[20] L. Gualtieri, E. Rauch, and R. Vidoni, "Emerging research fields in safety and ergonomics in industrial collaborative robotics: A systematic literature review," Robotics and Computer-Integrated Manufacturing, vol. 67, 2021, Art. no. 101998, doi: 10.1016/j.rcim.2020.101998.

Authorized licensed use limited to: Lomonosov Moscow State University. Downloaded on October 12,2026 at 07:12:57 UTC from IEEE Xplore. Restrictions apply.